

C++ List

List in C ++ programming language is a dynamic container which allows insertion and deletion at any location.

C++ List Declaration

```
list < datatype > list_Name
```

For example,

```
list < int > myList,
```

List of functions associated with Stack

- push_front(element):
- push_back(element):
- pop_front():
- pop_back():
- size():
- empty():

- `sort()`:
 - `unique()`:
 - `clear()`:
 - `reverse ()`:
-

Example1: List in C++

```
#include <iostream>
#include <list>
using namespace std;

void listPrint ( list < int > );

int main()
{
    list <int> myList;
    myList.push_front(100);
    myList.push_front(200);
    myList.push_back(50);
    myList.push_back(30);

    myList.insert(myList.begin(), 3, 300);

    //myList.sort();
    //myList.unique();
    //myList.clear();
    //myList.reverse();
    //myList.size();
    cout<<"the size is " <<myList.size()<<endl;
    listPrint(myList);
    return 0;
}

void listPrint ( list < int > list1) // Function to display all the list elements
{
    cout<<"list Items are:";
    while ( !list1.empty() )           //while list is not empty
    {
        cout << '\t' << list1.front();    // print the front item then delete it (to avoid endless loop)
        list1.pop_front();
    }
}
```

```
the size is 7
list Items are: 300    300    300    200    100    50    30
```

Exercise:

- 1) Practice the different built-in functions of List, and compare them to Stack and Queue functions.